

# Project Overview: NeuroFleetX - AI-Driven Urban Mobility Optimization System



## AI-Based System

Fleet and traffic optimization using intelligent routing algorithms and predictive analytics



## Comprehensive Features

Routing, real-time tracking, maintenance scheduling, and booking management in unified platform



## Internship Project

Built during Infosys Springboard program with mentorship and agile development practices

# System Architecture



## Frontend Layer

React dashboards with responsive UI components and real-time data visualization



## Backend APIs

Spring Boot services handling business logic, authentication, and data processing



## Data Storage

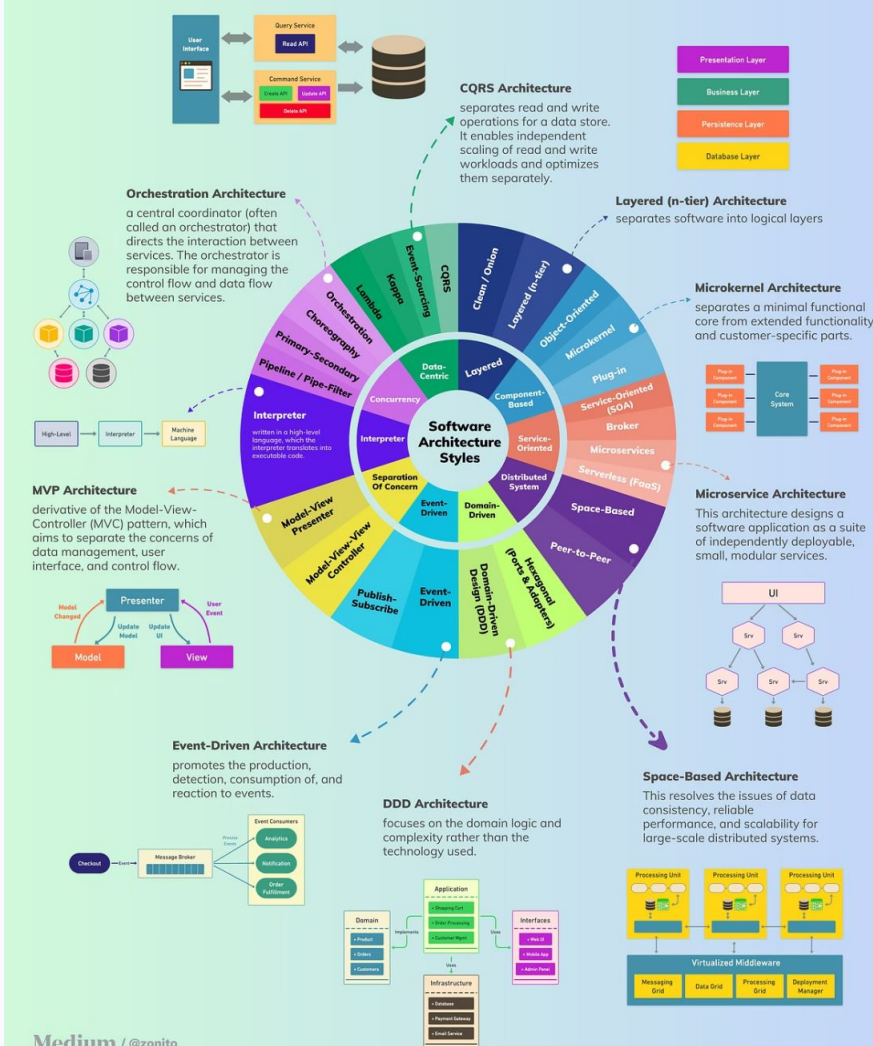
Postgres SQL relational database for fleet data, routes, bookings, and maintenance records



## AI Engine

Intelligent algorithms for route optimization, traffic prediction, and maintenance alerts

## Software Architecture Styles



# Modules Implemented

## Core System Components

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### Authentication & Role Management

Secure user authentication with role-based access control for drivers, fleet managers, and administrators

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### Fleet Management System

Vehicle tracking, status monitoring, and fleet composition management with real-time GPS data integration

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### Route Optimization Module

Dynamic path calculation using shortest path algorithms with traffic-aware routing logic

# Modules



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## Predictive Maintenance

Planned predictive maintenance based on vehicle usage data, mileage thresholds, and rule-based analysis

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## Booking & Scheduling

Reservation system with time slot management, vehicle availability checks, and booking confirmation workflows

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## Admin Dashboard

Dashboard interface displaying route history, vehicle status, and basic operational insights

# Key Features Developed

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## 1 Real-Time Vehicle Tracking

Simulated real-time vehicle tracking using GPS-like data, with dynamic location updates and status indicators (active, idle, maintenance)

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## 2 Dynamic Route Calculation

Dynamic route calculation using algorithmic optimization and heuristic-based adjustments based on vehicle availability

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## 3 Maintenance Alert System

Maintenance alert system based on usage thresholds and rule-based analysis for timely notifications

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## 4 Dashboard Analytics

Dashboard visualizations displaying route history, vehicle status, and basic operational insights

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## 5 Booking Interface

Proposed booking interface for managing reservations and vehicle availability

# AI & Algorithmic Logic

## Dijkstra Algorithm

Implemented shortest path calculation for optimal route finding between multiple waypoints with weighted edges

## Traffic-Aware Routing

Simulated traffic-aware routing using dynamic weighting logic for route optimization

## Predictive Maintenance

Planned predictive maintenance using data-driven analysis of vehicle usage and wear patterns

## Smart Recommendations

Rule-based smart recommendations for vehicle allocation, route selection, and scheduling

# System Workflow

- The system works in a continuous flow, starting from vehicle data inputs and ending with dashboard outputs
- Simulated GPS, vehicle status, and sensor data are collected to mimic real-time conditions
- This data is transmitted to the backend through API requests for further processing
- The backend processes, validates, and securely stores the data in the Postgres database
- AI logic is applied to optimize routes and predict maintenance requirements
- Based on analysis, optimal routes and alert notifications are generated
- Finally, the processed information is displayed on the dashboard as maps, analytics, and vehicle status

# Challenges Faced



## Module Integration

Coordinating frontend React components with Spring Boot backend services and AI algorithm integration required careful interface design



## Data

- Used dummy data to simulate real-time vehicle tracking
- Managed multiple vehicle updates in parallel
- Maintained consistent data flow in the system



## API Communication

- Debugged asynchronous API calls between frontend and backend
- Handled error responses and improved reliability
- Ensured smooth data transmission across system components



## Security Issues

Implementing robust authentication middleware and role-based access control without compromising system performance



## System Scalability

Designing database schemas and API structures that can accommodate growing fleet sizes and increasing data volume

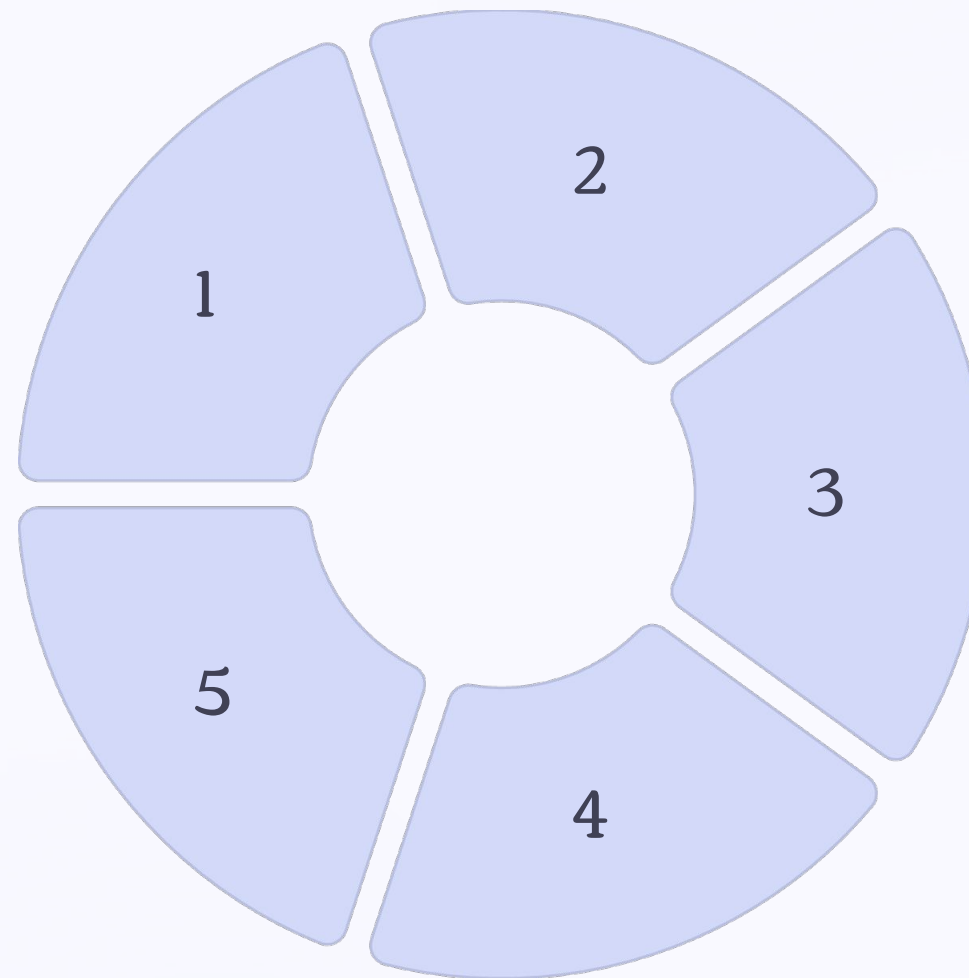
# Key Learnings

## Full-Stack Development

Hands-on experience building complete applications from React frontend through Spring Boot backend to Postgres persistence

## Agile Collaboration

Daily standups, sprint planning, and version control workflows that mirror professional software development practices



## API Integration & Testing

Skills in RESTful API design, endpoint testing with Postman, and debugging cross-module communication

## Database Design

Relational database modeling, schema optimization, query performance tuning, and data integrity constraints

## AI Implementation

Applied algorithmic thinking to real-world problems, balancing computational efficiency with accuracy requirements

# Outcome & Future Scope

-  **Working Prototype**  
Successfully developed functional system with all core modules integrated and tested end-to-end
-  **Technical Mastery**  
Deepened understanding of smart mobility systems architecture and real-time fleet management challenges
-  **Future Enhancements**  
IoT sensor integration for real vehicle telemetry, live traffic API connections, machine learning models for predictive analytics
-  **Mobile Application**  
Cross-platform mobile app for drivers and passengers with push notifications and location-based services

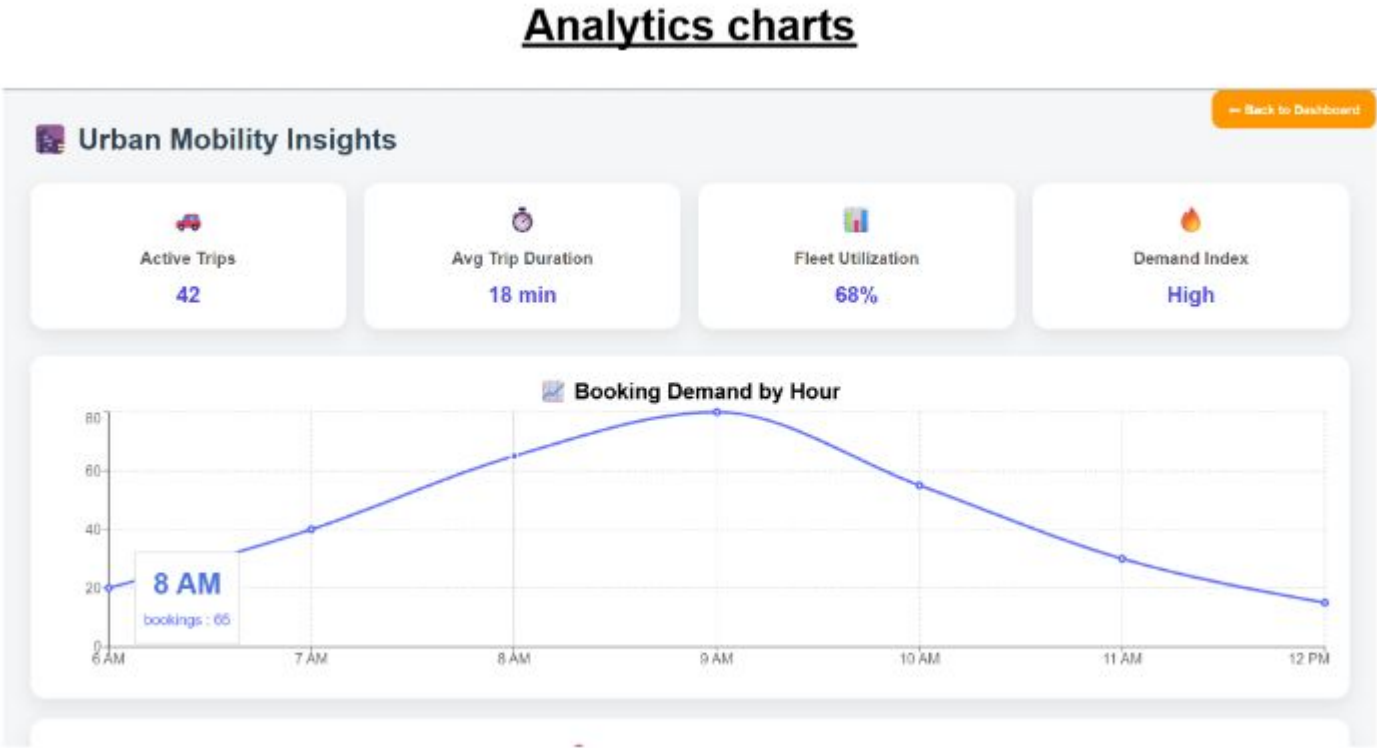
System Implementation Screens

### Register

Customers

Register

Already have an account? [Login](#)



| Fleet Vehicles |              |            |          |               |              |      |             |        |
|----------------|--------------|------------|----------|---------------|--------------|------|-------------|--------|
| ID             | Number       | Type       | Capacity | Model         | Manufacturer | Year | Status      | Action |
| 5              | TN01BC2345   | Car        | 3000     | Hyundai verna | Hyundai      | 2025 | Available   | Delete |
| 6              | TN01TRK1001  | Mini Truck | 1000     | Tata Ace      | Tata         | 2023 | Available   | Delete |
| 7              | TN01VAN2001  | Van        | 500      | Bolero Pickup | Mahindra     | 2021 | Available   | Delete |
| 8              | TN01MINI3001 | Mini Van   | 300      | Eeco          | Maruti       | 2022 | Available   | Delete |
| 4              | TN01AB1234   | Truck      | 5000     | Tata 407      | Tata         | 2022 | Maintenance | Delete |
| 3              | TN01AB1234   | Truck      | 5000     | Tata 407      | Tata         | 2022 | In Use      | Delete |
| 9              | TN01BC1234   | car        | 500      | Vista         | TATA         | 2021 | In Use      | Delete |

